TIØ4145 Corporate Finance Overview Chapter 1-5, and 7

Maria Lavrutich
Email: maria.lavrutich@ntnu.no
Tel: 90402416
Web: www.ntnu.no/ansatte/maria.lavrutich

Department of Industrial Economics and Technology Management
Norwegian University of Science and Technology
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TIØ4145 Corporate Finance

- Chapter 1: The Corporation
- Chapter 2: Financial Statement Analysis
- Chapter 3: No-Arbitrage, Law of One Price
- Chapter 4: Time Value of Money
- Chapter 5: Interest Rates
- Chapter 7: Investment Decision Rules

These chapters are covered in the courses TIØ4258 Technology Management, TIØ4116 Microeconomics and Investment Analysis, and TIØ4105 Management Accounting.

I briefly repeat/list the concepts here. Students without the background need to examine the literature carefully as “self-study”.

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Chapter 1: The Corporation

- The Types of Firms
  - Sole Proprietorships
  - Partnerships
  - Limited Liability Companies
  - Corporations
- Features of Corporations
- Organizational chart of a typical corporation

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Chapter 1: The Corporation

- Ownership Versus Control of Corporations
  - The role of the corporate management team, equity holders and board of directors, CEO and CFO
  - The financial manager role (Investment, financing, risk management, treasury, tax/accounting)
  - The goal of the firm and society, ethics and incentives within corporations
  - Takeovers and bankruptcies

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Chapter 1: The Corporation

- The Stock Market
  - Private and Public companies
  - Primary versus Secondary markets
  - Exchange versus OTC markets

- Financial Markets
  - Stock Markets
  - Interest and Bond Markets
  - Currency Markets
  - Commodity Markets (Energy, Metals, Agriculture, etc.)

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Chapter 2: Financial Statement Analysis

- Firms Disclosure of Financial Information
- The Balance Sheet
  - Current and Non-Current Assets
  - Current and Non-Current Liabilities
  - Shareholders Equity
  - Market to Book Ratio
  - Enterprise Value

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Financial Statement Analysis

- **The Income Statement**
  - Gross Profit
  - Operating Expenses
  - Earnings Before Interest and Taxes (EBIT)
  - Pretax and Net Income
  - Earnings Per Share (EPS)

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Financial Statement Analysis

- **The Statement of Cash Flow**
  - Cash from operating activities
  - Cash from investment activities
  - Cash from financing activities
  - Foreign exchange activities
  - Change in cash and cash equivalents
- **Changes in Shareholders Equity**

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Key Financial Ratios

Profitability Ratios	
Gross Margin	$\frac{\text{Gross Profit}}{\text{Sales}}$
Operating Margin	$\frac{\text{Operating Income}}{\text{Sales}}$
EBIT Margin	$\frac{\text{EBIT}}{\text{Sales}}$
Net Profit Margin	$\frac{\text{Net Income}}{\text{Sales}}$
Liquidity Ratios	
Current Ratio	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$
Quick Ratio	$\frac{\text{Cash \& Short-term Investments} + \text{Accounts Receivable}}{\text{Current Liabilities}}$
Cash Ratio	$\frac{\text{Cash}}{\text{Current Liabilities}}$
Working Capital Ratios	
Accounts Receivable Days	$\frac{\text{Accounts Receivable}}{\text{Average Daily Sales}}$
Accounts Payable Days	$\frac{\text{Accounts Payable}}{\text{Average Daily Cost of Sales}}$
Inventory Days	$\frac{\text{Inventory}}{\text{Average Daily Cost of Sales}}$
Accounts Receivable Turnover	$\frac{\text{Annual Sales}}{\text{Accounts Receivable}}$
Accounts Payable Turnover	$\frac{\text{Annual Cost of Sales}}{\text{Accounts Payable}}$
Inventory Turnover	$\frac{\text{Annual Cost of Sales}}{\text{Inventory}}$

Interest Coverage Ratios	
EBIT/Interest Coverage	$\frac{\text{EBIT}}{\text{Interest Expense}}$
EBITDA/Interest Coverage	$\frac{\text{EBITDA}}{\text{Interest Expense}}$
Leverage Ratios	
Debt-Equity Ratio	$\frac{\text{Total Debt}}{\text{Book (or Market) Value of Equity}}$
Debt-to-Capital Ratio	$\frac{\text{Total Debt}}{\text{Total Equity} + \text{Total Debt}}$
Debt-to-Enterprise Value Ratio	$\frac{\text{Net Debt}}{\text{Enterprise Value}}$
Equity Multiplier (book)	$\frac{\text{Total Assets}}{\text{Book Value of Equity}}$
Equity Multiplier (market)	$\frac{\text{Enterprise Value}}{\text{Market Value of Equity}}$
Valuation Ratios	
Market-to-Book Ratio	$\frac{\text{Market Value of Equity}}{\text{Book Value of Equity}}$
Price-Earnings Ratio	$\frac{\text{Share Price}}{\text{Earnings per Share}}$
Enterprise Value Ratios	$\frac{\text{Enterprise Value}}{\text{EBIT or EBITDA or Sales}}$
Operating Returns	
Asset Turnover	$\frac{\text{Sales}}{\text{Total Assets}}$
Return on Equity (ROE)	$\frac{\text{Net Income}}{\text{Book Value of Equity}}$
Return on Assets (ROA)	$\frac{\text{Net Income} + \text{Interest Expense}}{\text{Book Value of Assets}}$
Return on Invested Capital (ROIC)	$\frac{\text{EBIT}\ (1 - \text{Tax Rate})}{\text{Book Value of Equity} + \text{Net Debt}}$

Financial Statement Analysis

- **(Big) Financial Reporting Failures**
  - Enron
  - WorldCom
  - Legislations following: Sarbanes Oxley Act (SOX) and Dodd Frank Act

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Chapter 3: Financial Decision Making and the Law of One Price

- Valuing Decisions
- Interest Rates and the Time Value of Money
- Present Value and the NPV Decision Rule
- Arbitrage and the Law of One Price
- No-Arbitrage and Security Prices
- Appendix: Risk Aversion and Risk Premium

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Valuation Principle

- **Central principle of finance:**
  - We can use current market prices to determine the value today of the costs and benefits associated with a decision.
  - Allows to apply the concept of NPV as a way to compare costs and benefits of a project in terms of a common unit.

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Chapter 4: The time Value of Money

- Valuing Cash Flows
- Calculating the Net Present Value
- Perpetuities and Annuities
- Using Excel Spreadsheets
- Non-Annual Cash Flows
- Solving for the Cash Payments
- The Internal Rate of Return

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Three Rules of Time Travel

Rule 1	Only values at the same point in time can be compared or combined.	
Rule 2	To move a cash flow forward in time, you must compound it.	Future Value of a Cash Flow $FV_n = C \times (1 + r)^n$
Rule 3	To move a cash flow backward in time, you must discount it.	Present Value of a Cash Flow $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$

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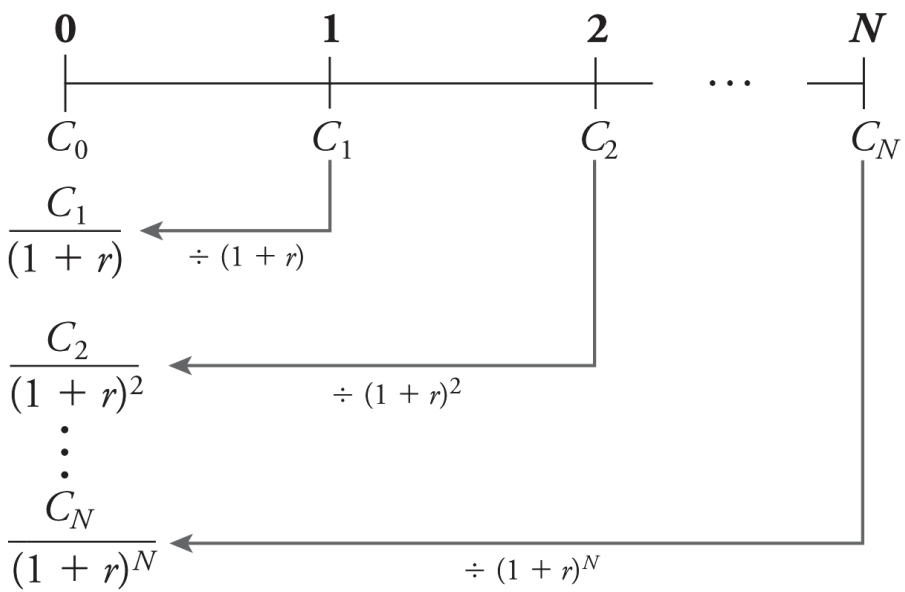
Valuing a Stream of Cash



$$C_0 + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \ldots + \frac{C_N}{(1+r)^N}$$

- Present Value of a Cash Flow Stream

$$PV = \sum_{n=0}^{N} PV(C_n) = \sum_{n=0}^{N} \frac{C_n}{(1+r)^n}$$

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Calculating the Net Present Value

- Calculating the NPV of future cash flows allows us to evaluate an investment decision.

- Net Present Value compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).

Perpetuities

- The value of a perpetuity (an amount of cash received forever) is simply the cash flow divided by the interest rate.

- Present Value of a Perpetuity

$$PV(C \text{ in perpetuity}) = \frac{C}{r}$$

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Annuities

- When a constant cash flow will occur at regular intervals for a finite number of N periods, it is called an *annuity*.

- Present Value of an Annuity

$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \ldots + \frac{C}{(1+r)^N} = \sum_{n=1}^{N} \frac{C}{(1+r)^n}$$

$$PV = C\frac{1}{r}\left(1 - \frac{1}{(1+r)^N}\right)$$

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The Internal Rate of Return

- In some situations, you know the present value and cash flows of an investment opportunity but you do not know *the internal rate of return (IRR)*

- **IRR** = the interest rate that sets the net present value of the cash flows equal to zero.

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Chapter 4: Interest Rates

- Interest Rate Quotes and Adjustments
- Application: Discount Rates and Loans
- The Determinants of Interest Rates
- Risk and Taxes
- The Opportunity Cost of Capital
- Continuous Rates and Cash Flows

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Interest Rate Quotes and Adjustments

Effective annual rates (EAR) for a 6% annual percentage rate (APR):

Compounding Interval	Effective Annual Rate
Annual	$(1 + 0.06/1)^1 - 1 = 6\%$
Semiannual	$(1 + 0.06/2)^2 - 1 = 6.09\%$
Monthly	$(1 + 0.06/12)^{12} - 1 = 6.1678\%$
Daily	$(1 + 0.06/365)^{365} - 1 = 6.1831\%$

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The Determinants of Interest Rates

- Inflation and Real versus Nominal Rates
  - Nominal Interest Rate: The rates quoted by financial institutions and used for discounting or compounding cash flows
  - Real interest Rate: The rate of growth of your purchasing power, after adjusting for inflation

$$r_r = \frac{r - i}{1 + i} \approx r - i$$

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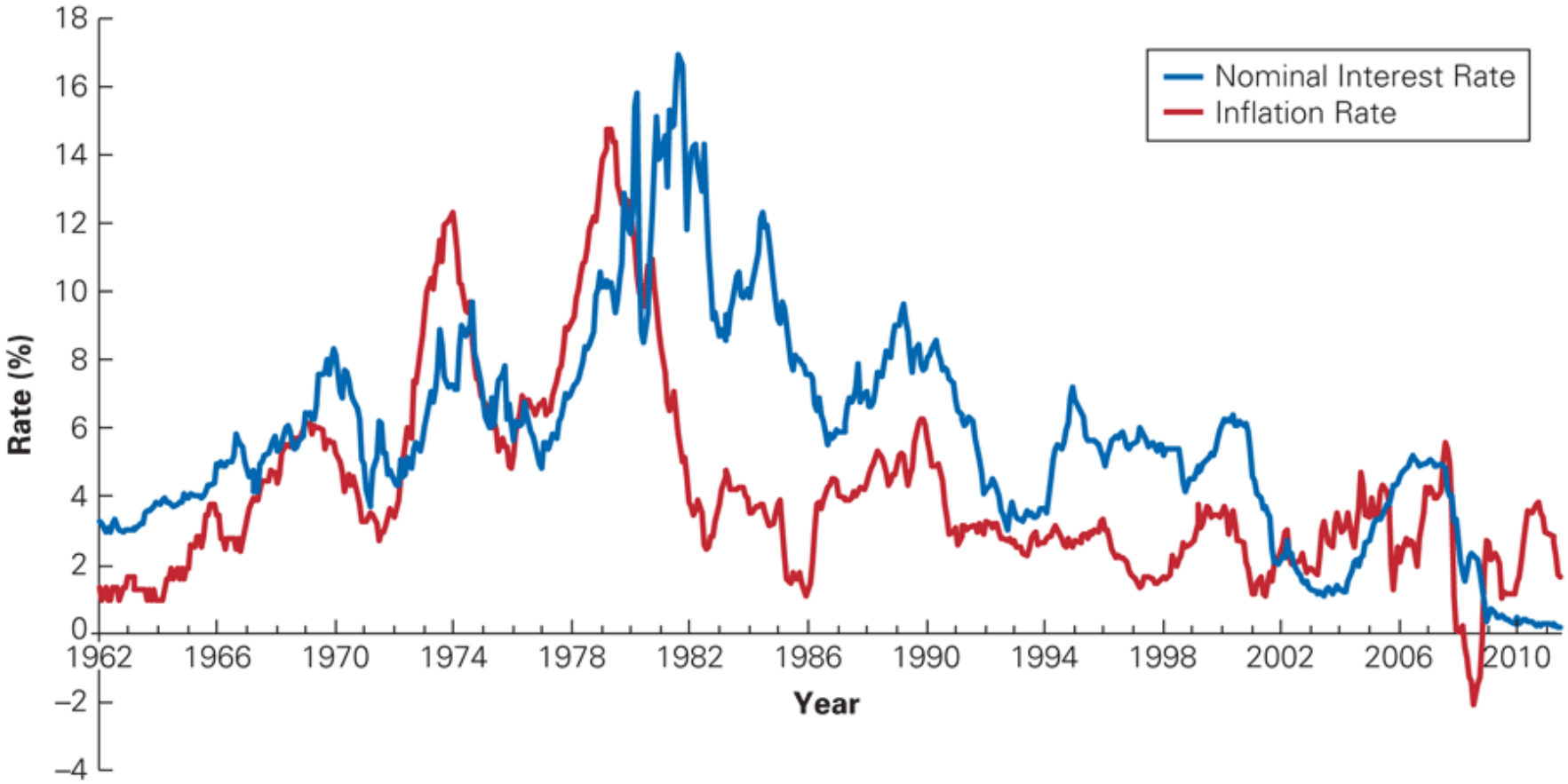
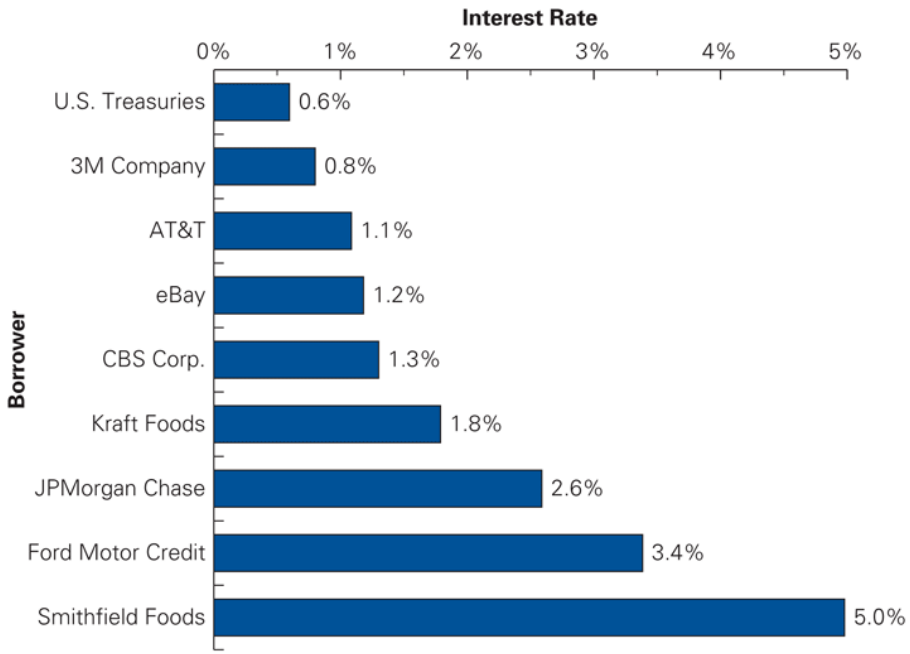
U.S. Interest Rates and Inflation Rates, 1960–2012



Risk and Taxes

- Risk and Interest Rates
  - U.S. Treasury securities are considered “risk-free.” All other borrowers have some risk of default, so investors require a higher rate of return on corporate bonds.



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After-Tax Interest Rates

- Taxes reduce the amount of interest an investor can keep, and we refer to this reduced amount as the **after-tax interest rate**.

$$r - (\tau \times r) = r\left(1 + \tau\right)$$

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The Opportunity Cost of Capital

- Investor's Opportunity Cost of Capital: The best available expected return offered in the market on an investment of comparable risk and term to the cash flow being discounted
  - Also referred to as *Cost of Capital*

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Continuous Rates and Cash Flows

- Continuously Arriving Cash Flows
  - If cash flows arrive, starting immediately, at an initial rate of $C per year, and if the cash flows grow at rate g per year, then given a discount rate of r per year, the present value of the cash flows is:
    - **Present Value of a Continuously Growing Perpetuity**

$$PV = \frac{C}{r_{cc} - g_{cc}}$$

    - where $r_{cc}=ln(1+r)$ and $g_{cc} = ln(1+g)$ are the discount and growth rates expressed as continuously APRs, respectively.

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Chapter 7: Investment Decision Rules

- NPV and Stand-Alone Projects
- The Internal Rate of Return Rule
- The Payback Rule
- Choosing Between Projects
- Project Selection with Resource Constraints

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Some problems with NPV and IRR

Problems with the IRR Rule

Problem

Consider projects with the following cash flows:

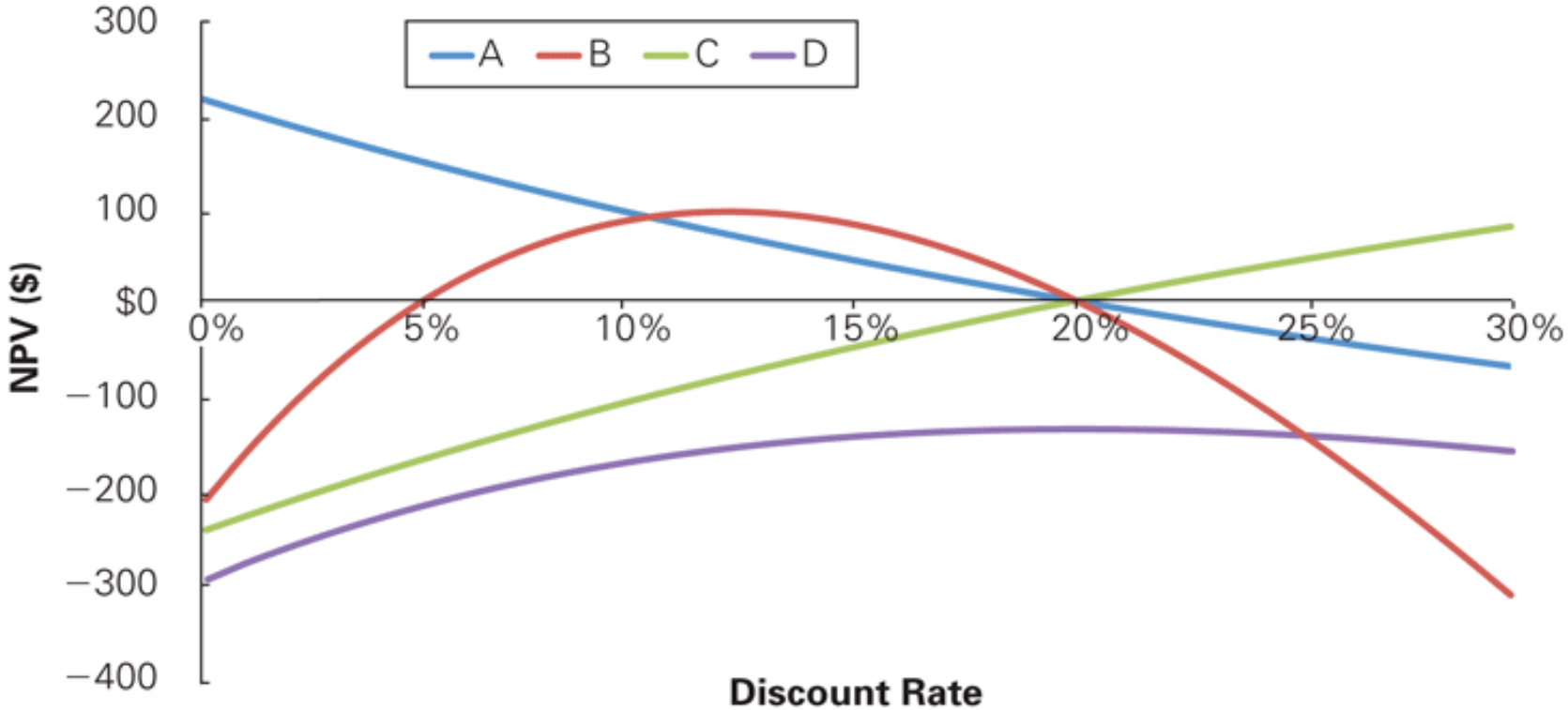
Project	0	1	2
A	−375	−300	900
B	−22,222	50,000	−28,000
C	400	400	−1,056
D	−4,300	10,000	−6,000

Which of these projects have an IRR close to 20%? For which of these projects is the IRR rule valid?

Solution

We plot the NPV profile for each project in Figure 7.5. From the NPV profiles, we can see that projects A, B, and C each have an IRR of approximately 20%, while project D has no IRR. Note also that project B has another IRR of 5%.

The IRR rule is valid only if the project has a positive NPV for every discount rate below the IRR. Thus, the IRR rule is only valid for project A. This project is the only one for which all the negative cash flows precede the positive ones.



Choosing Between Projects

- Mutually Exclusive Projects
  - When you must choose only one project among several possible projects, the choice is mutually exclusive.
  - NPV Rule
    - Select the project with the highest NPV.
  - IRR Rule
    - Selecting the project with the highest IRR may lead to mistakes.

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- Read textbook
- Read slides, make own notes
- Do selected exercises (both small examples and usage of Excel)
- Ask teacher/students assistance of background for the course.

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